

Ishwarya Anandakrishnan

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SUMMARY

AI/ML Engineer specializing in Generative AI, machine learning, and cloud-native applications. Expertise in developing production-grade AI platforms, LLM-powered applications, and ML infrastructure, leveraging RAG, NLP, MLOps, distributed systems, and backend engineering to deliver scalable, high-impact intelligent solutions.

EDUCATION

NC State University - <i>Master of Science, Computer Science</i> GPA: 3.7/4.0	Aug 2024 - May 2026
NIT Trichy - <i>Bachelor of Technology, Computer Science & Engineering</i> GPA: 8.5/10.0	Jul 2019 - May 2023

PROFESSIONAL EXPERIENCE

AI/ML Engineer, Cloud Pharmaceuticals Durham, NC (Remote)	Jun 2026 - Present
- Engineering a modular AI platform integrating unstructured EMR, genomic, and wearable data using NLP, LLMs, and time-series ML, reducing multimodal patient data preparation from 45 minutes to under 6 minutes for downstream medication optimization. - Architecting a scalable, resilient, and observable Generative AI fusion framework that synthesizes clinical, genomic, and physiological signals to accelerate evidence retrieval and enable explainable drug discovery and real-world evidence workflows.	

Research Assistant, NC State University Raleigh, NC	Jan 2026 - May 2026
- Developed a production GenAI Q&A platform using RAG, LangChain, LangGraph, GPT-4, and MongoDB Atlas, serving 50+ researchers across 1,000+ unstructured documents with sub-3-second response latency. - Established a RAG evaluation framework using RAGAS to benchmark faithfulness, context recall, and hallucination; applied preference optimization via human feedback to align model outputs, improving retrieval precision by 12% across 500+ evaluation queries. - Built scalable ML inference infrastructure on AWS ECS using Docker, Kubernetes, Prometheus, and Grafana, automating model deployment, autoscaling, and production monitoring through CI/CD pipelines that reduced deployment time by 70%.	

Software Engineer, Strand Life Sciences Bangalore, India	Jul 2023 - Jul 2024
- Engineered distributed REST APIs and ETL pipelines using Python, Java, Terraform, and AWS, processing 8B+ genomic records while reducing pipeline runtime from 10 hours to 7 hours through parallelized microservices. - Developed NLP entity-extraction models using spaCy and Hugging Face, supporting three production ML services while standardizing model evaluation using F1, Precision-Recall, and AUC benchmarks for production deployment. - Automated reporting workflows using Python and Google APIs, saving 4+ manual hours daily while enabling near real-time report generation to support faster model retraining across cross-functional teams.	

Undergraduate Researcher, NIT Trichy Trichy, India	Aug 2021 - May 2023
- Implemented Deep Q-Network (DQN) reinforcement learning agents for dynamic network optimization, improving network throughput by 15%, automating retraining through CI/CD, and presenting the research at IC@MACE 2023. - Created a non-contact body measurement system using OpenCV, Convex Hull, and 3D-scanned datasets, automating anthropometric measurements and earning Best Presentation among 70 competing teams at the Pragyan Tech Event.	

Software Engineering Intern, Strand Life Sciences Bangalore, India	May 2022 - Jul 2022
- Engineered a CAS-based Single Sign-On (SSO) solution integrated with Google OAuth, securing access for 7+ internal applications while streamlining centralized authentication and user lifecycle management. - Developed automated integration and regression test suites while implementing ELK Stack logging and observability, reducing incident MTTR by 20% and accelerating root-cause analysis for engineering teams.	

PROJECTS

Local GenAI File System Assistant github	Mar 2026
Built a local LLM over a Qdrant vector store enabling on-device semantic file search, content Q&A, and natural-language file operations via a full RAG pipeline; implemented a retrieval eval harness (precision@k, MRR) benchmarking chunking and embedding strategies	

Model Evaluation: Uncertainty Quantification Framework github	Jan 2026
Built a model evaluation and calibration framework using conformal prediction and MC Dropout across regression and classification tasks; achieved 90–95% empirical coverage; benchmarked ECE, calibration curves, and reliability diagrams over 5+ architectures	

Scalable ML Infrastructure: Kubernetes Deployment and CI/CD github	Nov 2025
Designed and deployed a production ML serving layer on Kubernetes with HPA autoscaling and Prometheus/Grafana monitoring; orchestrated 10+ experiment runs via MLflow and Airflow; CI/CD cut redeployment steps 80%	

Clinical NLP Transformer: Flash Attention and Inference Optimization github	Oct 2024
Implemented a decoder-style transformer with custom tokenisation and Flash Attention achieving 2× throughput over standard attention; built model evaluation harness for NLP classification achieving 0.75 F1 versus a non-attention baseline model on Portuguese medical text	

SKILLS

Programming Languages : Python, Java, TypeScript, SQL, C++, Bash
Machine Learning & AI : PyTorch, TensorFlow, Scikit-learn, XGBoost, CNNs, Transformers, Reinforcement Learning (DQN), Time-Series ML, NLP, spaCy, Hugging Face, Conformal Prediction, MC Dropout, Flash Attention, Pandas, NumPy, Statistical Modeling
Generative AI & LLMs : LangChain, LangGraph, LlamaIndex, OpenAI GPT-4, Ollama, RAG, RAGAS, Prompt Engineering, Vector Search, PEFT, LoRA, QLoRA
Data & Vector Databases : MongoDB Atlas, PostgreSQL, Qdrant, FAISS, Data Mining, Data Management
Cloud & MLOps : Docker, Kubernetes, MLflow, Airflow, Terraform, CI/CD, Prometheus, Grafana, AWS (ECS, EC2, S3, Lambda), Ansible
Backend & APIs : FastAPI, REST APIs, ETL Pipelines, Google APIs
Developer Tools : Git, GitHub, ELK Stack
Certifications : AI Fluency (Anthropic), Generative AI (NVIDIA), Deep Learning (NVIDIA), Claude 101 (Anthropic)